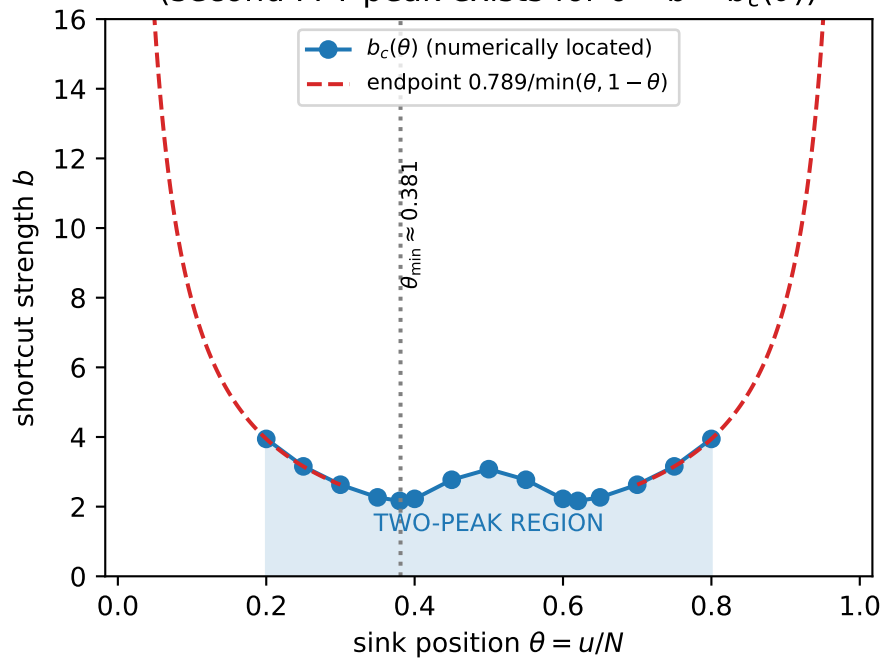
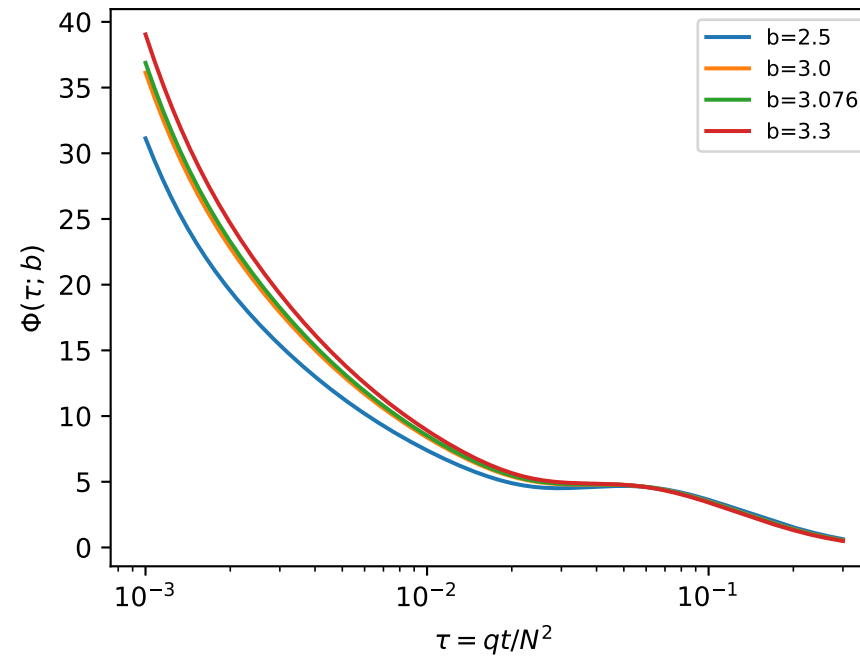


Directed-shortcut first-passage saddle-node: existence, universality (PRR figure suite)

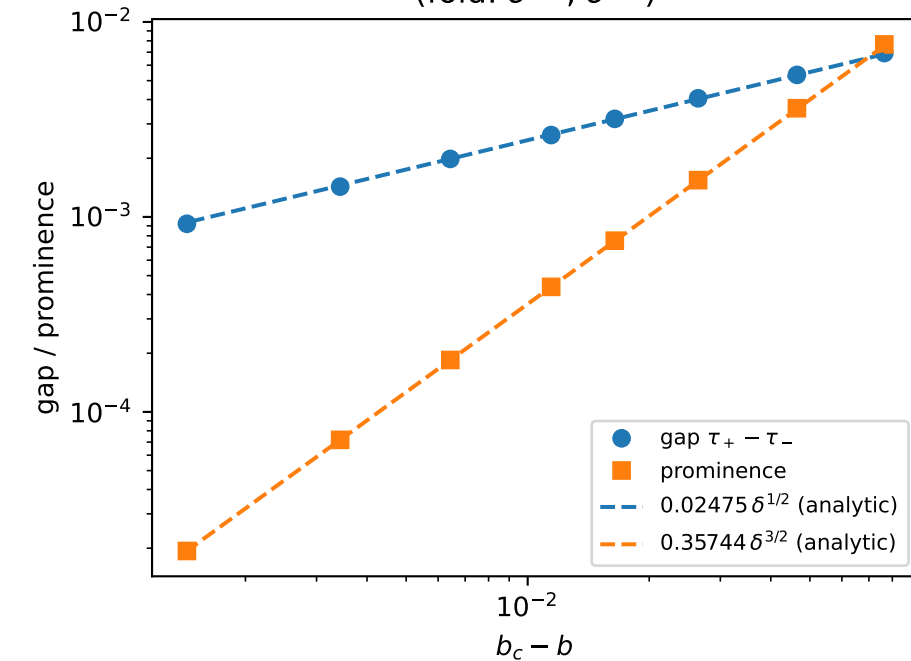
A. Saddle-node phase boundary
(second FPT peak exists for $0 < b < b_c(\theta)$)



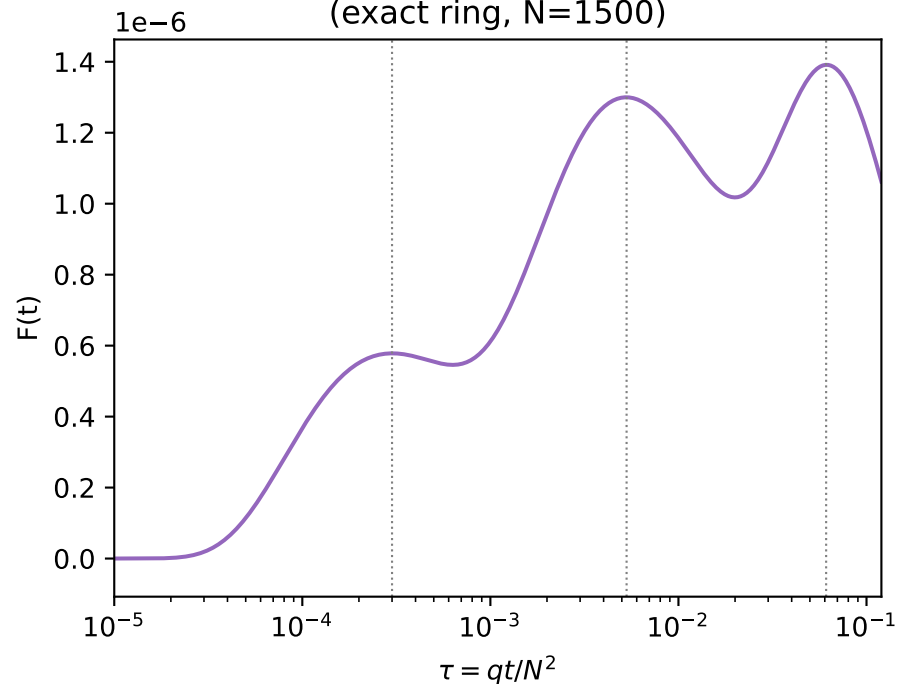
B. Master curve $\theta = 1/2$: second peak
annihilates at $b_c = 3.0764$



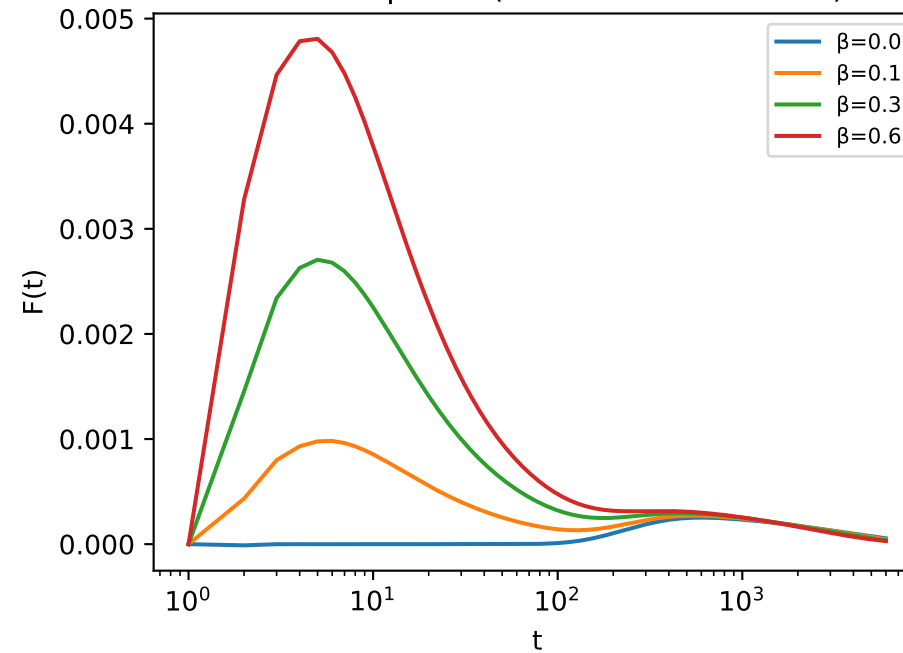
C. Saddle-node normal form
(fold: $\delta^{1/2}$, $\delta^{3/2}$)



D. Two shortcuts $\rightarrow$ THREE peaks
(exact ring, $N=1500$)



E. 2D torus (31×31): capture +
diffusive peaks (mechanism survives)



F. 2D fold: diffusive peak
prominence $\rightarrow 0$ at $\beta_c^{2D} \approx 0.69$

